Candidate's Seat no: ____________

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Sixth Semester of B. Tech. Examination (EE)
May 2013

EE308 High Voltage Engineering

Date: 02.05.2013, Thursday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Define Townsend's first and second ionization coefficients. How is the condition for breakdown obtained in Townsend's discharge? [07]

Q - 2 (a) Explain the mechanism of breakdown due to internal discharges occurred in solid dielectrics. [03]

(b) Explain breakdown phenomenon observed in commercial liquid dielectrics. [04]

(c) Explain the working of Cockcroft-Walton voltage multiplier circuit with neat sketch. [07]

OR

Q - 2 (a) Explain the mechanism of breakdown due to treeing and tracking in case of solid insulations. [06]

(b) Describe the dielectric properties of transformer oil as insulation. [02]

(c) What is the principle of operation of a resonant transformer? How it is advantageous over cascaded connected transformer? [06]

Q - 3 (a) A Cockcroft-Walton type voltage multiplier has eight stages with capacitances, all equal to 0.05 μF. The supply transformer secondary voltage is 125 kV at a frequency of 150 Hz. If the load current to be supplied is 5 mA, find (a) the percentage ripple, (b) the regulation, and (c) the optimum number of stages for minimum regulation or voltage drop. [07]

(b) An impulse generator has eight stages with each condenser rated for 0.16 μF and 125 kV. The load capacitor available is 1000 pF. Find the wave front resistance and the wave tail resistance needed to produce 1.2/50 μs impulse wave. [04]

(c) A generating voltmeter has to be designed so that it can have range from 20 to 200 kV dc. If indicating meter reads a minimum current of 2 μA and maximum current of 25 μA, what should the capacitance of generating voltmeter be? Assume that the driving motor has a synchronous speed of 1500 rpm. [03]

OR

Q - 3 (a) Enlist the methods used for the generation of high voltages and explain van de Graaff generator in detail. [07]

(b) Explain trigatron gap in detail. [05]

(c) Define wave front time and wave tail time. [02]

SECTION – II

Q - 4 (a) Enlist the methods used for the measurement of high voltages and explain any one method in detail. [07]

Q - 5 (a) Enlist the methods used for the measurement of high currents and explain any one in detail. [07]

(b) Describe with suitable diagram, the sphere gap arrangement and discuss the factors influencing the voltage measurement. [07]

OR

Q - 5 (a) Explain high voltage Schering bridge for measurement of capacitance and dielectric loss factor. [07]

(b) Define followings: [07]

[a] Dielectric loss [b] Corona discharge

[c] Flashover [d] Partial discharge

[e] Withstand voltage [f] Dissipation factor

[g] Disruptive discharge voltage

Q - 6 Explain any two from the followings in detail: [14]

1. Synthetic testing on circuit breaker
2. Testing on insulators
3. Lay out of high voltage laboratory
4. Testing on bushing

****